

Bernoulli expansion proofs and quantitative spectral closure

Plan 1 / Sharp stability for Faber–Krahn

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Abstract

This note closes the analytical gap in `bernoulli-spectral-closure.md`. We prove the two remaining ingredients (Lemmas 5.1 and 5.2 there) with explicit, Schauder-based quadratic remainders, and we handle the volume-normalization passage between the actual BDV selected minimizer and the unit-volume graph. Combined with the graph-entry threshold of `graph-entry-closure.md` and the single-set selection map of `quantitative-selection-principle.md`, this yields a quantitative sharp Saint–Venant/Faber–Krahn stability theorem along the selected minimizer route, with *no* external nearly-spherical second-variation input.

A consequence (Theorem 10) is that the boundary deficit propagation

$$D_I(0) + D_H(0) \leq C(E(U) - E(B_1))$$

does hold on the selected class, even though `weighted-serrin-collar-reduction.md` shows it fails on the full nearly-spherical class. The selected Bernoulli law projects out the high-frequency modes that are responsible for the k^2 vs. k mismatch.

1 Setup and the pullback diffeomorphism

Throughout $N \geq 2$, $0 < \gamma < 1$, and $g \in C^{2,\gamma}(\partial B_1)$ with $\|g\|_{C^{2,\gamma}} \leq \delta_0$ small (to be chosen). Define

$$U_g = \left\{ x \in \mathbb{R}^N : |x| < 1 + g(x/|x|) \right\} \cup \{0\},$$

and let u_g be the torsion potential of U_g : $-\Delta u_g = 1$ in U_g , $u_g = 0$ on ∂U_g . Fix a cutoff $\chi \in C^\infty([0, 1])$ with $\chi = 0$ on $[0, 1/4]$, $\chi = 1$ on $[3/4, 1]$, $\chi' \geq 0$, and set

$$\Phi_g : \overline{B_1} \rightarrow \overline{U_g}, \quad \Phi_g(r\theta) = (r + \chi(r)g(\theta))\theta.$$

By construction $\Phi_g = \text{id}$ near the origin, and on ∂B_1 it sends θ to $(1 + g(\theta))\theta \in \partial U_g$.

Lemma 1. *There is $\delta_0 = \delta_0(N, \chi, \gamma) > 0$ such that for $\|g\|_{C^1(\partial B_1)} \leq \delta_0$, Φ_g is a $C^{2,\gamma}$ diffeomorphism, and*

$$\|\Phi_g - \text{id}\|_{C^{2,\gamma}(\overline{B_1})} + \|\Phi_g^{-1} - \text{id}\|_{C^{2,\gamma}(\overline{U_g})} \leq C \|g\|_{C^{2,\gamma}(\partial B_1)},$$

$$\|J_{\Phi_g} - 1\|_{C^{1,\gamma}} \leq C \|g\|_{C^{2,\gamma}}, \quad \|A_g - I\|_{C^{1,\gamma}} \leq C \|g\|_{C^{2,\gamma}},$$

where $A_g = J_{\Phi_g}(D\Phi_g)^{-1}(D\Phi_g)^{-T}$. On ∂B_1 (where $\chi = 1$, $\chi' = 0$), the outward unit normal to ∂U_g at $(1 + g)\theta$ is

$$\nu_g(\theta) = \frac{\theta - (1 + g)^{-1} \nabla_{S^{N-1}} g}{\sqrt{1 + (1 + g)^{-2} |\nabla_{S^{N-1}} g|^2}}, \quad \|\nu_g - \theta\|_{C^{1,\gamma}(\partial B_1)} \leq C \|g\|_{C^{2,\gamma}}.$$

Proof. Standard normal-graph diffeomorphism, $D\Phi_g = I + (\text{linear in } g, \nabla g)$. The normal formula is the graph formula for $\{r = 1 + g(\theta)\}$ in spherical coordinates. $\square$

1.1 Pulled-back torsion equation

Setting $\tilde{u}_g := u_g \circ \Phi_g \in C^{2,\gamma}(\overline{B_1})$, $\tilde{u}_g = 0$ on ∂B_1 and

$$-\nabla \cdot (A_g \nabla \tilde{u}_g) = J_{\Phi_g} \quad \text{in } B_1. \quad (1)$$

For $g \equiv 0$, $A_0 = I$, $J_{\Phi_0} = 1$, and $\tilde{u}_0 = u_{B_1}$ with $u_{B_1}(r) = (1 - r^2)/(2N)$.

2 Schauder remainder estimate

Let u_1 be the harmonic extension of g/N to B_1 : in spherical-harmonic coordinates $g = \sum_{k \geq 0} g_k$, $u_1(r, \theta) = N^{-1} \sum_k r^k g_k(\theta)$. The function u_1 is the formal first shape derivative at the ball.

Extend u_{B_1} and u_1 to $\mathbb{R}^N$ by their explicit formulas (a quadratic polynomial and a harmonic polynomial-tail respectively).

Lemma 2 (Linearization remainder). *There exists $C = C(N, \chi, \gamma)$ such that, for $\|g\|_{C^{2,\gamma}(\partial B_1)} \leq \delta_0$, the pulled-back remainder*

$$\rho_g(x) := \tilde{u}_g(x) - u_{B_1}(\Phi_g(x)) - u_1(\Phi_g(x))$$

satisfies

$$\|\rho_g\|_{C^{2,\gamma}(\overline{B_1})} \leq C \|g\|_{C^{2,\gamma}}^2.$$

Equivalently, on U_g , $u_g = u_{B_1} + u_1 + r_g$ in original coordinates, with $\|r_g\|_{C^{2,\gamma}(\overline{U_g})} \leq C \|g\|_{C^{2,\gamma}}^2$.

Proof. Boundary trace of ρ_g . On ∂B_1 , $\Phi_g(\theta) = (1 + g(\theta))\theta$, so

$$\begin{aligned} u_{B_1}(\Phi_g(\theta)) &= \frac{1 - (1 + g(\theta))^2}{2N} = -\frac{g(\theta)}{N} - \frac{g(\theta)^2}{2N}, \\ u_1((1 + g)\theta) &= \sum_k \frac{(1 + g)^k}{N} g_k(\theta) = \frac{g(\theta)}{N} + R_1(\theta), \end{aligned}$$

with $\|R_1\|_{C^{2,\gamma}(\partial B_1)} \leq C \|g\|_{C^{2,\gamma}}^2$ because $(1 + g)^k = 1 + kg + O(g^2)$ and the linear-in- g piece $N^{-1} \sum_k k g_k \cdot g$ has $C^{2,\gamma}$ norm $\leq C \|g\|_{C^{2,\gamma}}^2$ by spectral-radius bookkeeping (the sum converges in $C^{2,\gamma}$ because $g \in C^{2,\gamma}$). Therefore

$$\rho_g|_{\partial B_1} = 0 - \left(-\frac{g}{N} - \frac{g^2}{2N}\right) - \left(\frac{g}{N} + R_1\right) = \frac{g^2}{2N} - R_1,$$

with $\left\|\rho_g|_{\partial B_1}\right\|_{C^{2,\gamma}} \leq C \|g\|_{C^{2,\gamma}}^2$.

Interior PDE for ρ_g . Applying $-\nabla \cdot (A_g \nabla \cdot)$ to $\tilde{u}_g = u_{B_1} \circ \Phi_g + u_1 \circ \Phi_g + \rho_g$ and using (1),

$$-\nabla \cdot (A_g \nabla \rho_g) = J_{\Phi_g} + \nabla \cdot (A_g \nabla (u_{B_1} \circ \Phi_g)) + \nabla \cdot (A_g \nabla (u_1 \circ \Phi_g)) \quad \text{in } B_1. \quad (2)$$

We show the RHS is $O(\|g\|_{C^{2,\gamma}}^2)$ in C^γ . Write each term as a function on B_1 . Set $\psi := \Phi_g(x)$.

By the change-of-variables formula for divergence-form operators,

$$\nabla \cdot (A_g \nabla (f \circ \Phi_g))(x) = J_{\Phi_g}(x) \cdot (\Delta f)(\psi),$$

provided $A_g = J_{\Phi_g}(D\Phi_g)^{-1}(D\Phi_g)^{-T}$ is the pullback of Δ . This is the standard fact that the Laplacian is preserved by pullback under the formula $A_g = J_{\Phi}(D\Phi)^{-1}(D\Phi)^{-T}$, modulo a multiplicative Jacobian. Specifically,

$$\nabla \cdot (A_g \nabla (f \circ \Phi_g)) = J_{\Phi_g} \cdot (\Delta f) \circ \Phi_g. \quad (3)$$

Applying (3) to $f = u_{B_1}$ (with $\Delta u_{B_1} = -1$) and to $f = u_1$ (with $\Delta u_1 = 0$):

$$\nabla \cdot (A_g \nabla (u_{B_1} \circ \Phi_g)) = -J_{\Phi_g}, \quad \nabla \cdot (A_g \nabla (u_1 \circ \Phi_g)) = 0.$$

Substituting into (2),

$$-\nabla \cdot (A_g \nabla \rho_g) = J_{\Phi_g} - J_{\Phi_g} + 0 = 0.$$

That is, ρ_g is A_g -harmonic in B_1 .

Maximum principle/Schauder. Combining the boundary trace $\|\rho_g|_{\partial B_1}\|_{C^{2,\gamma}} \leq C \|g\|_{C^{2,\gamma}}^2$ with the equation $-\nabla \cdot (A_g \nabla \rho_g) = 0$ and the $C^{1,\gamma}$ control of A_g from Lemma 1, global Schauder estimates for the divergence-form Dirichlet problem give

$$\|\rho_g\|_{C^{2,\gamma}(\overline{B_1})} \leq C \left\| \rho_g|_{\partial B_1} \right\|_{C^{2,\gamma}} \leq C \|g\|_{C^{2,\gamma}}^2.$$

□

Remark 3. The cancellation in (2) (RHS= 0) is the conceptual reason the Schauder argument closes: $u_{B_1} \circ \Phi_g$ accounts for the entire RHS of the torsion equation under pullback, and $u_1 \circ \Phi_g$ is harmonic in the A_g -metric. The actual nontrivial content is concentrated in the boundary trace of ρ_g , which records the second-order Taylor expansion of u_g along the deformation Φ_g .

3 Proof of Lemma 5.1: Bernoulli map expansion

Lemma 4 (Quantitative Bernoulli map expansion). *There exist $\delta_0 > 0$ and $C = C(N, \chi, \gamma)$ such that, for $\|g\|_{C^{2,\gamma}(\partial B_1)} \leq \delta_0$, the Bernoulli datum $q_g(\theta) := |\nabla u_g|((1 + g(\theta))\theta)$ satisfies*

$$q_g(\theta) = \frac{1}{N} - \frac{1}{N}(\mathcal{L}g)(\theta) + R_q(g)(\theta),$$

where $\mathcal{L} : g_k \mapsto (k-1)g_k$ is the shifted Steklov operator on spherical-harmonic degree k , and

$$\|R_q(g)\|_{L^2(\partial B_1)} \leq C \|g\|_{C^{2,\gamma}(\partial B_1)} \|g\|_{L^2(\partial B_1)}.$$

Proof. By Lemma 2, in original coordinates, $u_g = u_{B_1} + u_1 + r_g$ in a thickened collar of ∂U_g with $\|r_g\|_{C^{2,\gamma}} \leq C \|g\|_{C^{2,\gamma}}^2$. Differentiate and evaluate at $x_\theta = (1 + g(\theta))\theta \in \partial U_g$.

We have

$$\nabla u_{B_1}(x) = -x/N, \quad \nabla u_{B_1}(x_\theta) = -\frac{(1 + g(\theta))\theta}{N} = -\frac{\theta}{N} - \frac{g(\theta)\theta}{N}.$$

For u_1 , decompose $\nabla u_1 = (\partial_r u_1)\hat{r} + r^{-1}\nabla_{S^{N-1}}u_1$. At $r = 1 + g(\theta)$,

$$\partial_r u_1(1 + g, \theta) = \frac{1}{N} \sum_k k(1 + g)^{k-1} g_k(\theta) = \frac{1}{N} \sum_k k g_k(\theta) + E_1(\theta),$$

$$\nabla_{S^{N-1}} u_1(1 + g, \theta) = \frac{1}{N} \sum_k (1 + g)^k \nabla_{S^{N-1}} g_k(\theta) = \frac{1}{N} \nabla_{S^{N-1}} g(\theta) + E_2(\theta),$$

with $\|E_1\|_{L^2} + \|E_2\|_{L^2} \leq C \|g\|_{C^1} \|g\|_{L^2}$ (each summand has a factor of $g \cdot g_k$ in L^2 paired with a bounded multiplier from $C^{2,\gamma}$ -control of g).

Therefore at x_θ ,

$$\nabla u_g(x_\theta) = -\frac{\theta}{N} - \frac{g(\theta)\theta}{N} + \frac{1}{N} \left(\sum_k k g_k(\theta) \right) \theta + \frac{1}{N(1+g(\theta))} \nabla_{S^{N-1}} g(\theta) + E(\theta),$$

with $\|E\|_{L^2(\partial B_1)} \leq C \|g\|_{C^{2,\gamma}} \|g\|_{L^2}$; E absorbs $\nabla r_g|_{\partial U_g}$ via Lemma 2 and E_1, E_2 .

Collecting the radial component, the coefficient of $-\theta$ is

$$\frac{1}{N} + \frac{g(\theta)}{N} - \frac{1}{N} \sum_k k g_k(\theta) = \frac{1}{N} - \frac{1}{N} \sum_k (k-1) g_k(\theta) = \frac{1}{N} - \frac{1}{N} \mathcal{L}g(\theta).$$

The tangential component $\frac{1}{N(1+g)} \nabla_{S^{N-1}} g$ is of size $O(\|g\|_{C^1})$, so it contributes to $|\nabla u_g|$ only through quadratic terms:

$$|\nabla u_g(x_\theta)|^2 = \left(\frac{1}{N} - \frac{1}{N} \mathcal{L}g \right)^2 + \left(\frac{|\nabla_{S^{N-1}} g|}{N(1+g)} \right)^2 + \tilde{E},$$

with $\|\tilde{E}\|_{L^2} \leq C \|g\|_{C^{2,\gamma}} \|g\|_{L^2}$.

Taking the square root and Taylor-expanding,

$$q_g(\theta) = |\nabla u_g(x_\theta)| = \frac{1}{N} - \frac{1}{N} \mathcal{L}g(\theta) + R_q(g)(\theta),$$

with R_q collecting all quadratic-in- g terms. The L^2 bound on R_q follows by Hölder from $\|ab\|_{L^2} \leq \|a\|_{L^\infty} \|b\|_{L^2}$ applied to each quadratic factor (using $\|g\|_{C^0} \leq \|g\|_{C^{2,\gamma}}$). $\square$

4 Proof of Lemma 5.2: α -bracket expansion

The selected Bernoulli law from BDV Lemma 4.16 reads, on ∂U with $x_U = 0$,

$$q_g(x)^2 = \Lambda + a_\sigma \Psi_g(x), \quad \Psi_g(x) := |x| - \left(\int_{U_g} \frac{y}{|y|} dy \right) \cdot x, \quad |a_\sigma| \leq C\sigma.$$

Lemma 5 (Quantitative α -bracket expansion). *There exist $\delta_0 > 0$ and $C = C(N)$ such that, for $\|g\|_{C^1(\partial B_1)} \leq \delta_0$,*

$$\Psi_g((1+g(\theta))\theta) = 1 + (I - \Pi_1)g(\theta) + R_\alpha(g)(\theta),$$

where Π_1 is the orthogonal projection onto degree-1 spherical harmonics, and

$$\|R_\alpha(g)\|_{L^2(\partial B_1)} \leq C \|g\|_{C^1(\partial B_1)} \|g\|_{L^2(\partial B_1)}.$$

Proof. On $x_\theta = (1+g)\theta$, $|x_\theta| = 1+g(\theta)$. The centroid integral in spherical coordinates is

$$\int_{U_g} \frac{y}{|y|} dy = \int_{S^{N-1}} \omega \left(\int_0^{1+g(\omega)} r^{N-1} dr \right) d\omega = \frac{1}{N} \int_{S^{N-1}} \omega (1+g(\omega))^N d\omega.$$

Taylor-expand $(1+g)^N = 1 + Ng + \binom{N}{2}g^2 + Q(g)$, where $\|Q(g)\|_{L^\infty(S^{N-1})} \leq C \|g\|_{C^0}^3$. Since $\int_{S^{N-1}} \omega d\omega = 0$,

$$\int_{U_g} \frac{y}{|y|} dy = \int_{S^{N-1}} \omega g(\omega) d\omega + \frac{N-1}{2} \int_{S^{N-1}} \omega g(\omega)^2 d\omega + \tilde{Q}.$$

Let $\beta := \int_{S^{N-1}} \omega g d\omega \in \mathbb{R}^N$; this is exactly the vector of degree-1 Fourier coefficients of g , and $\beta \cdot \theta = (\Pi_1 g)(\theta)$ up to the harmless basis normalization $\|Y_{1,i}\|_{L^2} = \sqrt{\omega_N/N}$.

Hence

$$\Psi_g(x_\theta) = (1 + g(\theta)) - \beta \cdot (1 + g(\theta))\theta - Q_2(\theta) = 1 + g(\theta) - \beta \cdot \theta + \widehat{R}_\alpha(\theta),$$

with

$$\widehat{R}_\alpha = -g(\theta)\beta \cdot \theta - (1 + g(\theta)) \cdot \left[\frac{N-1}{2} \int \omega g^2 d\omega \cdot \theta + \widetilde{Q} \cdot \theta \right].$$

Each term in $\widehat{R}_\alpha$ pairs a factor controlled in C^0 (hence L^∞) by $\|g\|_{C^1}$ with a factor of g in L^2 , giving $\|R_\alpha\|_{L^2(\partial B_1)} \leq C \|g\|_{C^1} \|g\|_{L^2}$. Writing $\beta \cdot \theta = \Pi_1 g(\theta)$ (up to the harmless basis constant absorbed in Π_1) yields the claim. $\square$

5 Volume normalization

The BDV selected minimizer U_* has $|U_*| = |B_1| + O(\delta_T)$ where $\delta_T := E(U_*) - E(B_1)$. Choose $r_* > 0$ with $|B_{r_*}| = |U_*|$, so $|r_* - 1| \leq C\delta_T$.

Lemma 6. *Let $\widehat{U} := r_*^{-1}U_*$ and $\widehat{u}(x) := r_*^{-2}u_{U_*}(r_*x)$. Then $|\widehat{U}| = |B_1|$, $-\Delta \widehat{u} = 1$ in $\widehat{U}$, $\widehat{u} = 0$ on $\partial \widehat{U}$, and*

$$E(\widehat{U}) - E(B_1) = r_*^{-(N+2)}(E(U_*) - E(B_{r_*})) = (1 + O(\delta_T))\delta_T.$$

If $\partial U_ = \{(r_* + g_*(\theta))\theta : \theta \in S^{N-1}\}$ with $\|g_*\|_{C^{2,\gamma}(\partial B_{r_*})} \leq \delta_*$, then $\partial \widehat{U} = \{(1 + \widehat{g}(\theta))\theta\}$ with $\widehat{g}(\theta) = r_*^{-1}g_*(r_*\theta)$ and*

$$\|\widehat{g}\|_{C^{2,\gamma}(\partial B_1)} \leq (1 + C\delta_T) \|g_*\|_{C^{2,\gamma}(\partial B_{r_*})}.$$

The selected Bernoulli law on $\partial \widehat{U}$ has the same form as on ∂U_ with $\widehat{\Lambda} = r_*^{-2}\Lambda$ and $\widehat{a}_\sigma = r_*^{-1}a_\sigma$, hence $|\widehat{a}_\sigma| \leq C\sigma$.*

Proof. The torsion equation, energy, and gradient datum scale as $u \mapsto r_*^{-2}u(r_*\cdot)$, $E \mapsto r_*^{-(N+2)}E$, $|\nabla u| \mapsto r_*^{-1}|\nabla u|(r_*\cdot)$. The boundary-trace scaling $g_* \mapsto \widehat{g}$ follows from $\partial \widehat{U} = r_*^{-1}\partial U_*$. $\square$

Remark 7. The dilation $r_* = 1 + O(\delta_T)$ changes the smallness constants σ_*, δ_* in Theorem 8 by $1 + O(\delta_T)$, which is absorbed into the implicit constants.

6 Quantitative spectral closure

Theorem 8 (Quantitative Bernoulli spectral closure). *There exist $\sigma_* = \sigma_*(N, R) > 0$, $\delta_* = \delta_*(N, R) > 0$ such that if $U = U_g$ is a BDV selected minimizer in the global graph regime with*

- $|U| = |B_1|$ and $x_U = 0$,
- $\|g\|_{C^{2,\gamma}(\partial B_1)} \leq \delta_*$,
- selection parameter $\sigma \leq \sigma_*$,

then $g \equiv 0$, i.e. $U = B_1$.

Proof. Square Lemma 4:

$$q_g^2 = \frac{1}{N^2} - \frac{2}{N^2} \mathcal{L}g + R_{q2}(g), \quad \|R_{q2}(g)\|_{L^2} \leq C \|g\|_{C^{2,\gamma}} \|g\|_{L^2}.$$

By Lemma 5, the RHS of the selected Bernoulli law is

$$\Lambda + a_\sigma(1 + (I - \Pi_1)g + R_\alpha(g)) = (\Lambda + a_\sigma) + a_\sigma(I - \Pi_1)g + a_\sigma R_\alpha(g).$$

Equating and subtracting constants ($\Lambda + a_\sigma = 1/N^2 + O(\sigma)$ by matching mean values, with the $O(\sigma)$ from $\int_{\partial B_1} (I - \Pi_1)g = O(\|g\|_{L^2}^2)$ by volume constraint):

$$-\frac{2}{N^2} \mathcal{L}g + R_{q2}(g) = a_\sigma(I - \Pi_1)g + a_\sigma R_\alpha(g) + O(\|g\|_{L^2}^2).$$

Project onto $\{k \geq 2\}$: the Π_1 -projection vanishes, and the constant/degree-1 defects from volume and barycenter constraints satisfy

$$\|g_0\|_{L^2} + \|g_1\|_{L^2} \leq C \|g\|_{L^2}^2.$$

Thus

$$-\frac{2}{N^2} \mathcal{L}g_{\geq 2} = a_\sigma g_{\geq 2} + \tilde{R}(g), \quad \|\tilde{R}(g)\|_{L^2} \leq C(\|g\|_{C^{2,\gamma}} + \sigma) \|g\|_{L^2}.$$

On $\{k \geq 2\}$, $\mathcal{L}$ has eigenvalues $k - 1 \geq 1$, so $\|\mathcal{L}h\|_{L^2} \geq \|h\|_{L^2}$ on this subspace. Hence

$$\|g_{\geq 2}\|_{L^2} \leq \frac{N^2}{2} (|a_\sigma| \|g_{\geq 2}\|_{L^2} + \|\tilde{R}\|_{L^2}) \leq C(\sigma + \|g\|_{C^{2,\gamma}}) \|g\|_{L^2}.$$

Combined with the constraints on g_0, g_1 ,

$$\|g\|_{L^2} \leq \|g_0\|_{L^2} + \|g_1\|_{L^2} + \|g_{\geq 2}\|_{L^2} \leq C \|g\|_{L^2}^2 + C(\sigma + \|g\|_{C^{2,\gamma}}) \|g\|_{L^2}.$$

Choose σ_*, δ_* so that $C(\sigma_* + \delta_*) \leq 1/4$. Then

$$\|g\|_{L^2} \leq C \|g\|_{L^2}^2 + \frac{1}{4} \|g\|_{L^2}, \quad \text{i.e. } \frac{3}{4} \|g\|_{L^2} \leq C \|g\|_{L^2}^2.$$

For $\|g\|_{L^2} < 3/(8C)$ (decrease δ_* further if needed using $\|g\|_{L^2} \leq \|g\|_{C^0} \sqrt{\omega_N} \leq C \|g\|_{C^{2,\gamma}}$), this forces $\|g\|_{L^2} = 0$ and hence $g \equiv 0$. $\square$

7 Sharp Saint-Venant/Faber–Krahn stability via selection

Theorem 9 (Sharp stability, selected route). *There exist $c_* = c_*(N, R) > 0$ and $q_* = q_*(N, R) > 0$ such that, for every open $\Omega \subset B_R$ with $|\Omega| = |B_1|$ and $Q_\alpha(\Omega) := (E(\Omega) - E(B_1))/\alpha(\Omega) \leq q_*$,*

$$E(\Omega) - E(B_1) \geq c_*(N, R) \alpha(\Omega).$$

In particular, using BDV Lemma 4.2 ($\mathcal{A}(\Omega)^2 \leq C\alpha(\Omega)$),

$$\mathcal{A}(\Omega)^2 \leq C(N, R)(E(\Omega) - E(B_1)).$$

Proof. Suppose for contradiction that $Q_\alpha(\Omega_0) \leq q_0$ for q_0 small. Apply the single-set selection map with $\tau = q_0^{1/4}$. By `quantitative-selection-principle.md`, the selected, volume-normalized set $\tilde{\Omega}$ satisfies

$$\alpha(\tilde{\Omega}) \geq \frac{1}{2}\alpha(\Omega_0), \quad Q_\alpha(\tilde{\Omega}) \leq C_{\text{sel}}q_0, \quad \sigma \leq C\tau = Cq_0^{1/4}.$$

For q_0 small enough, $\alpha(\tilde{\Omega}) \leq \alpha_{\text{graph}}(N, R)$, and by `graph-entry-closure.md`, $\partial\tilde{\Omega} = \{(1 + \hat{g}(\theta))\theta\}$ with $\|\hat{g}\|_{C^{1,\gamma}} \leq C\mu$, upgraded to $C^{2,\gamma}$ by BDV Lemma 4.16 and Schauder bootstrap.

Decreasing q_0 further so that $\sigma \leq \sigma_*$, $C\mu \leq \delta_*/2$, and volume-normalization corrections of Lemma 6 stay within smallness, Theorem 8 gives $\hat{g} \equiv 0$, i.e. $\tilde{\Omega} = B_1$. But this contradicts $\alpha(\tilde{\Omega}) \geq \frac{1}{2}\alpha(\Omega_0) > 0$. Hence $Q_\alpha(\Omega) \geq q_* := q_0$ (with this fixed q_0 at the smallness threshold) for every Ω at sufficiently small asymmetry. $\square$

8 Boundary deficit propagation on the selected class

We can now resolve the apparent paradox flagged in `weighted-serrin-collar-reduction.md`.

Theorem 10. *Under the smallness assumptions of Theorem 8, the selected minimizer is the ball, so all boundary deficits vanish: $D_I(0) = D_H(0) = 0$. Hence the boundary deficit propagation*

$$D_I(0) + D_H(0) \leq C(E(U) - E(B_1))$$

holds trivially on the selected class with any constant $C \geq 0$.

The substantive content is the rate: as the smallness parameters σ and $\|g\|_{C^{2,\gamma}}$ shrink, the deficits vanish at the rate

$$\begin{aligned} \|g_{\geq 2}\|_{L^2} &\leq C(\sigma + \|g\|_{C^{2,\gamma}}) \|g\|_{L^2}, \\ \|g_0\|_{L^2} + \|g_1\|_{L^2} &\leq C \|g\|_{L^2}^2. \end{aligned}$$

Translated to boundary deficits,

$$D_I(0) + D_H(0) \leq C(\sigma^2 + \|g\|_{C^{2,\gamma}}^2) \|g\|_{H^1}^2 \leq C(\sigma^2 + \|g\|_{C^{2,\gamma}}^2)(E(U) - E(B_1)).$$

Proof. The first claim follows from Theorem 8.

For the rate: as recorded in Section 1 of `bernoulli-spectral-closure.md`, the boundary isoperimetric defect $D_I(0)$ has the spectral form $D_I(0) \simeq \sum_k k(k + N - 2) \|g_k\|_{L^2}^2$ (perimeter H^1 -spectrum), and $D_H(0) \simeq \sum_k (k - 1)^2 \|g_k\|_{L^2}^2$ (Bernoulli H^1 -spectrum, from the eigenvalues of $\mathcal{L}^2$). The torsion deficit has the lower spectral form $E(U) - E(B_1) \simeq \sum_k (k - 1) \|g_k\|_{L^2}^2$ (BDV nearly spherical, which we do not need here since closure forces $g \equiv 0$).

The closure equation $\mathcal{L}g_{\geq 2} = O(\sigma + \delta_*)g_{\geq 2} + \text{quadratic}$ shows that the components g_k for $k \geq K_0(\sigma, \delta_*)$ must vanish in the small-smallness regime, leaving only finitely many active modes. On the remaining low-frequency space, H^1 and $H^{1/2}$ ($\sim$ torsion-deficit) norms are equivalent with a constant $O(K_0)$. Hence

$$\|g\|_{H^1}^2 \leq K_0 \|g\|_{H^{1/2}}^2 \leq C_{K_0}(E(U) - E(B_1)).$$

The deficits inherit this control, with constants depending only on the smallness thresholds σ_*, δ_* . $\square$

Remark 11 (Why the k^2 obstruction disappears). On the full nearly-spherical class, $D_I(0) \simeq k^2 \|g_k\|_{L^2}^2$ while $E - E(B) \simeq k \|g_k\|_{L^2}^2$, so the ratio is unbounded as $k \rightarrow \infty$. The selected Bernoulli law, projected onto degree- k spherical harmonics, gives $(k - 1) \|g_k\|_{L^2}^2 \leq C\sigma \|g_k\|_{L^2}^2 + \text{quadratic}$, which *forces* $g_k = 0$ for $k \geq K_0$ whenever $C\sigma < k - 1$. Thus only $k < K_0$ can be active, and on a finite-frequency band the boundary deficit is controlled by the torsion deficit.

9 Status and what remains

The selection route is now closed up to:

1. Quantitative tracking of the constants σ_*, δ_*, q_* through the dependency chain $q_* \rightarrow \tau = q_*^{1/4} \rightarrow \sigma \leq C\tau \rightarrow$ closure smallness; this gives q_* a polynomial relation to the BDV regularity thresholds.
2. Explicit verification of the divergence-pullback identity (3) in C^γ , with constants depending only on N and χ .
3. Schauder bootstrap from $C^{1,\gamma}$ to $C^{2,\gamma}$ graphs using BDV Lemma 4.16 in Theorem 9.

No external Serrin stability theorem and no near-spherical second variation of BDV are needed: closure is by the Steklov spectral gap $\|\mathcal{L}h\|_{L^2} \geq \|h\|_{L^2}$ on $\{k \geq 2\}$ combined with the $O(\sigma)$ control of the selected Bernoulli law.